

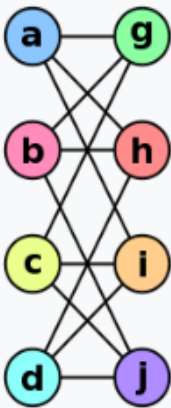
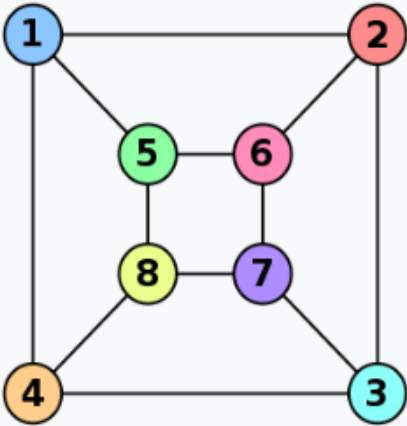
3. 3 Isomorphism of Graphs

In graph theory, an **isomorphism of graphs** G and H is a bijection between the vertex sets of G and H

$$f:V(G) \rightarrow V(H)$$

such that any two vertices u and v of G are adjacent in G if and only if $f(u)$ and $f(v)$ are adjacent in H . This kind of bijection is commonly described as "edge-preserving bijection", in accordance with the general notion of isomorphism being a structure-preserving bijection. If an isomorphism exists between two graphs, then the graphs are called **isomorphic** and denoted as $G \cong H$. In the case when the bijection is a mapping of a graph onto itself, i.e., when G and H are one and the same graph, the bijection is called an automorphism of G . If a graph is finite, we can prove it to be bijective by showing it is one-one/onto; no need to show both. Graph isomorphism is an equivalence relation on graphs and as such it partitions the class of all graphs into equivalence classes. A set of graphs isomorphic to each other is called an **isomorphism class** of graphs. The question of whether graph isomorphism can be determined in polynomial time is a major unsolved problem in computer science, known as the Graph Isomorphism problem.

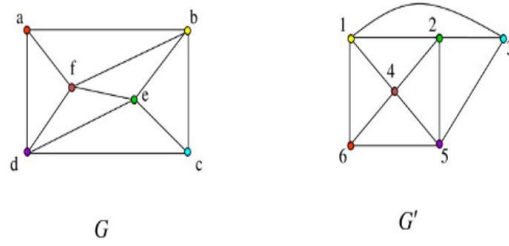
The two graphs shown below are isomorphic, despite their different looking drawings

Graph G	Graph H	An isomorphism between G and H
		$f(a) = 1$ $f(b) = 6$ $f(c) = 8$ $f(d) = 3$ $f(g) = 5$ $f(h) = 2$ $f(i) = 4$ $f(j) = 7$

Exercise 3.3 Isomorphism of Graphs

Directions: Determine the isomorphism between the 2 given graphs.

A.



B.

